MUHAMMAD JUNAID

Professional Portfolio Technical Report

Computer Science Department | COMSATS University Islamabad

October 2026

Live Website: https://port-inky-ten.vercel.app

GitHub Repository: https://github.com/junaid-devv/portfolio

Project Overview

This report documents the personal portfolio website developed for Muhammad Junaid, a Computer Science undergraduate at COMSATS University Islamabad. The portfolio features a modern, dark aesthetic interface engineered with HTML5, CSS3, and vanilla JavaScript. It demonstrates mastery across full-stack web development, mobile application creation, AI agent integration, and 2D mobile game engineering.

The web application is fully responsive across mobile, tablet, and desktop viewports, and has been deployed live to Vercel with source code hosted publicly on GitHub.

Interface Preview & Visuals

Hero & Landing Section

The landing viewport features dynamic typewriter text rotation, real-time status badges, interactive stats counters, and a 3D perspective avatar with floating skill badges.



Figure 1 -- Hero Section with Dynamic Typing and Floating Badges

Technical Skills

Skills categorized into Web Development, App Development, and Core Engineering, equipped with scroll-triggered animated progress indicators.



Figure 2 -- Skills and Technologies Matrix

About & Academic Journey

Narrative cards highlighting background, software vision, and current projects alongside an education timeline covering BSCS coursework at COMSATS.

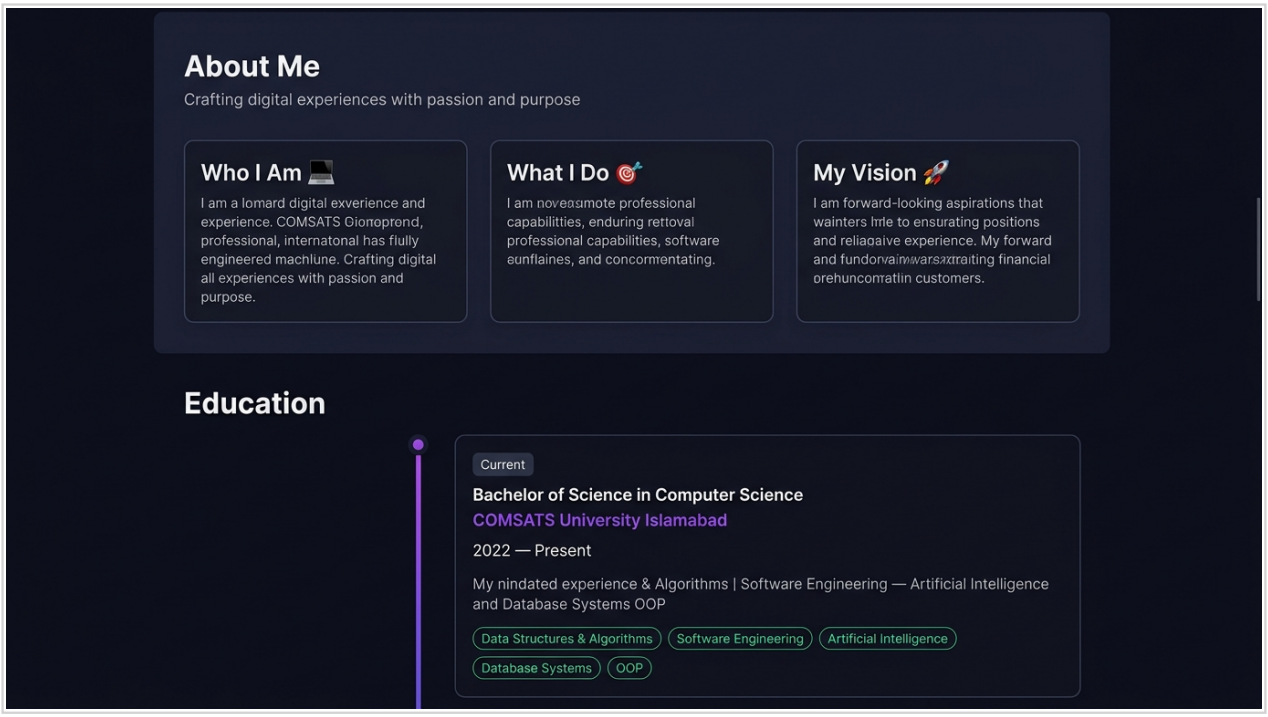


Figure 3 -- About Me and University Education Timeline

Interactive JavaScript Operations

To provide an engaging, responsive user experience without heavy external dependencies, the portfolio embeds native vanilla JavaScript operations:

1. Interactive Accordion Component

An expandable technical breakdown and FAQ accordion in the Deep-Dive section:

- Smooth height transition using dynamic scrollHeight calculation.
- Single-item focus pattern: opening an item collapses any open sibling item.
- Accessible markup with dynamic aria-expanded state and 180-degree rotating chevrons.
- Detailed breakdowns for Campus Mate, WhatsApp AI Agent, and 2D Game architecture.

2. Featured Projects Showcase Slideshow (Carousel)

An interactive project carousel highlighting flagship achievements:

- Horizontal sliding track with smooth cubic-bezier transitions.
- Next and Previous circular navigation controls with glassmorphism backdrop.
- Interactive pagination dot indicators with active pulse-stretch state.
- Dynamic numeric slide counter (e.g. 01 / 03).
- Automatic slideshow interval (5.5s) with pause-on-hover and mobile touch-swipe support.

3. Text & Visual Animations

Multiple coordinated animations enhancing visual hierarchy and interaction feedback:

- Dynamic Typewriter Effect: Cycles through 'Full Stack Developer', 'Mobile App Developer', 'AI Solutions Builder', and '2D Game Developer' with blinking neon cursor.
- Scroll Reveal Animations: Elements slide and fade into view using IntersectionObserver API.
- Animated Number Counters: Stats count up dynamically from 0 to target metrics upon scroll.
- 3D Card Tilt: Micro-interaction tilting cards dynamically based on cursor coordinates.
- Circular Progress Back-to-Top Button: SVG ring fills proportionately with page scroll depth.

Responsive Architecture

The layout is engineered using mobile-first CSS media queries ensuring seamless responsiveness:

- Desktop (> 1024px): 3-column grids, horizontal navigation links, side-by-side hero layout.
- Tablet (768px - 1024px): 2-column adaptive grids with adjusted padding and spacing.
- Mobile (< 768px): Animated hamburger button toggles an off-canvas drawer navigation menu with frosted glass backdrop; hero visual and content stack vertically for optimal touch navigation.

Technical Skills Matrix

Web Development

Skill	Proficiency
HTML5 & Modern Semantic Web	90%
CSS3, Flexbox, Grid, Animations	88%
JavaScript (ES6+) & DOM APIs	82%
React.js & Single Page Apps	75%
Node.js & Express REST APIs	72%

Mobile App Development

Skill	Proficiency
Flutter & Dart	82%
Firebase & Cloud Firestore	80%
RESTful API Integration	85%
State Management & Architecture	76%
UI/UX Prototyping & Design	75%

Engineering & Tools

Skill	Proficiency
Git, GitHub & Version Control	88%
Python & AI Integration	80%
C++ & Object-Oriented Design	78%
Databases (SQL & NoSQL)	78%
Game Development (Unity 2D)	70%

Featured Projects Summary

1. Campus Mate App (Flagship Mobile App)

Comprehensive mobile companion for university students. Features automated timetable alerts, interactive campus navigation, weighted GPA calculator, and real-time push announcements.

Status	Completed
Tech Stack	Flutter, Dart, Firebase Firestore, Cloud Messaging, REST APIs

2. WhatsApp AI Agent for Restaurants (AI & Automation)

Autonomous conversational assistant processing incoming webhooks via WhatsApp Business API. Uses natural language processing to handle menu browsing, food ordering, and table reservations.

Status	Completed
Tech Stack	Python, WhatsApp API, NLP Intent Parser, Node.js, MongoDB

3. 2D Mobile Game (Game Development)

Physics-driven 2D action game featuring responsive touch controls, hand-crafted sprite animations, modular level obstacles, particle effects, and dynamic high-score tracking.

Status	In Active Development
Tech Stack	Unity, C# Scripting, 2D Rigidbody Physics, Sprite Animation

Academic Background

Degree	Bachelor of Science in Computer Science (BSCS)
University	COMSATS University Islamabad
Duration	2022 - Present (Current)

Core coursework: Data Structures & Algorithms, Object-Oriented Analysis & Design, Database Management Systems, Operating Systems, Computer Architecture, and Artificial Intelligence.

Interests & Pursuits

- Competitive programming and algorithmic problem solving (LeetCode).
- Modern UI/UX prototyping and micro-interaction engineering.
- Open-source software collaboration and repository maintenance.
- Game mechanics design, sprite physics, and interactive entertainment.
- Machine learning applications, NLP agents, and workflow automation.

Contact Information

Channel	Details
Email	junaid@example.com
GitHub	github.com/junaid-devv
LinkedIn	linkedin.com/in/muhammadjunaid
Location	Islamabad, Pakistan